ABV IIITM STUDENT BRANCH

IEEE MP SECTION

HACKSAGON

A NATIONAL LEVEL SOFTWARE & HARDWARE HACKATHON

AgroWise & AgroSense

Bridging the Digital Divide with Phygital Precision Agriculture.

TEAM NAMOFANS

SOFTWARE + HARDWARE

Track: Agritech & Rural Innovation

Objective: To empower Indian farmers with a ‘Phygital’ (Physical + Digital) ecosystem that combines ground-truth soil telemetry with accessible, vernacular AI. We aim to ensure precision agriculture is available to everyone, from smartphone users to feature phone owners.

Sustainable Development Goals Addressed

1 NO POVERTY

Goal 1: No Poverty. Reduces input waste, **saving** ₹50K+/hectare.

15 LIFE ON LAND

Goal 15: Life on Land. Cuts chemical overuse by 40%, regenerating soil health.

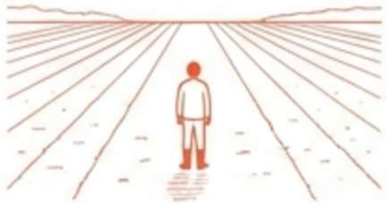
2 ZERO HUNGER

Goal 2: Zero Hunger. Prevents crop failure, securing food supply.

PROBLEM STATEMENT

The 'Invisible' Soil Crisis

Critical stress factors like root-zone moisture deficits and pH imbalances are invisible to the naked eye. Farmers currently rely on lagging visual cues like wilting. By the time a plant wilts, physiological damage is permanent, leading to a reactive cycle of over-watering (root rot) or under-watering (stunting).



Farming in the Dark

There is a massive information gap. Generic weather apps provide district-level averages that miss critical microclimates (e.g., fungal risk risk in a humid field). With only ~1 extension officer per 1,000+ farmers, scientific soil testing is a logistical luxury most never receive.

The Digital Divide (Accessibility Barrier)

The majority of Agritech solutions are 'Smartphone-First' or require 4G. This design choice alienates the most vulnerable demographic—smallholder farmers who rely on feature phones or live in areas with patchy connectivity—leaving them behind in the precision agriculture revolution.

PROPOSED SOLUTION

1. AgroSense (Hardware)

The Ground Truth

A solar-powered, 'deploy-and-forget' IoT spike. It sits in the root zone, measuring soil moisture, temperature, and humidity. It requires no buttons or battery swaps (solar + deep sleep), ensuring autonomous operation.

2. AgroWise PWA (Software)

The High-Tech Layer

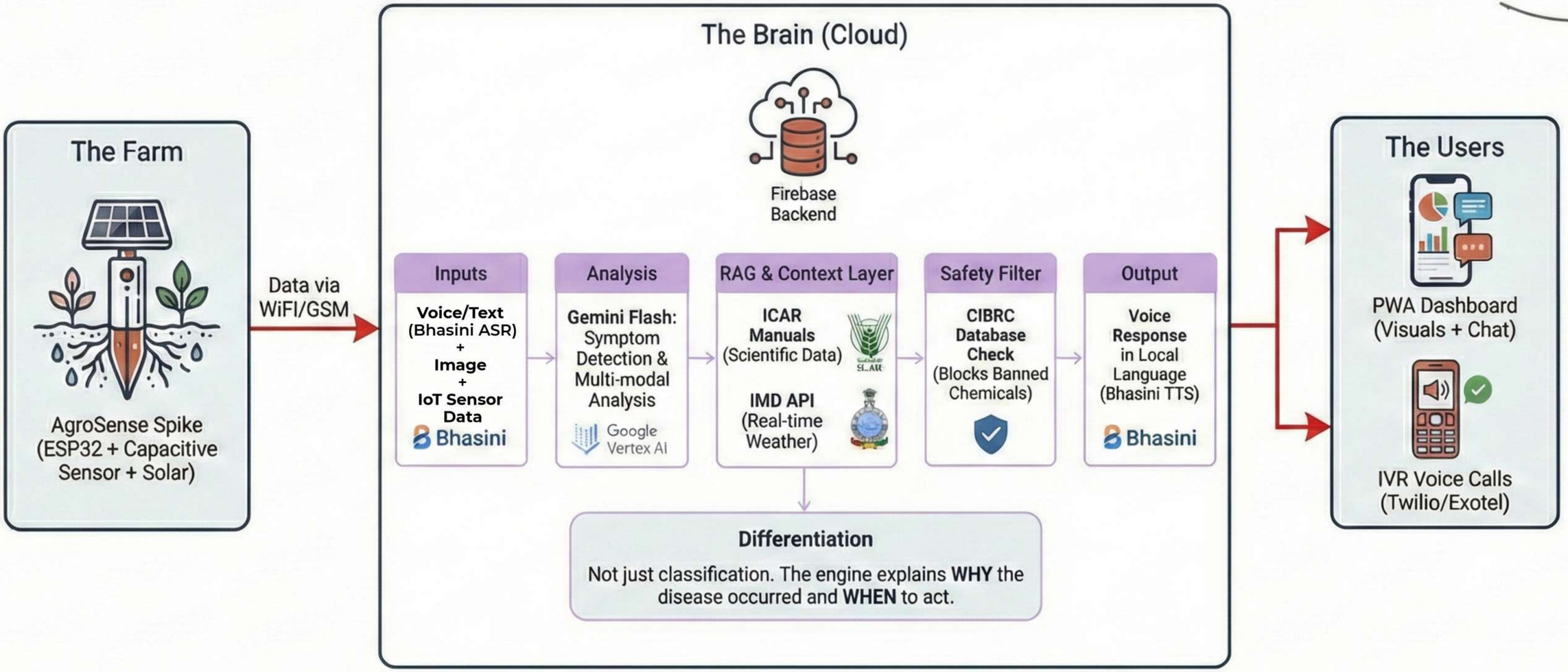
For smartphone users, this Progressive Web App provides a visual dashboard. It features 'Soil Thirst' gauges and uses 'Neuro-Symbolic' AI to diagnose plant diseases by correlating leaf photos with soil sensor data.

3. Kisan-Vani (Voice Interface)

The High-Reach Layer

A toll-free IVR for feature phones. It bridges the digital divide by calling farmers with voice alerts in their local dialect when critical thresholds are met, and allows them to 'dial their farm' to check status without internet.

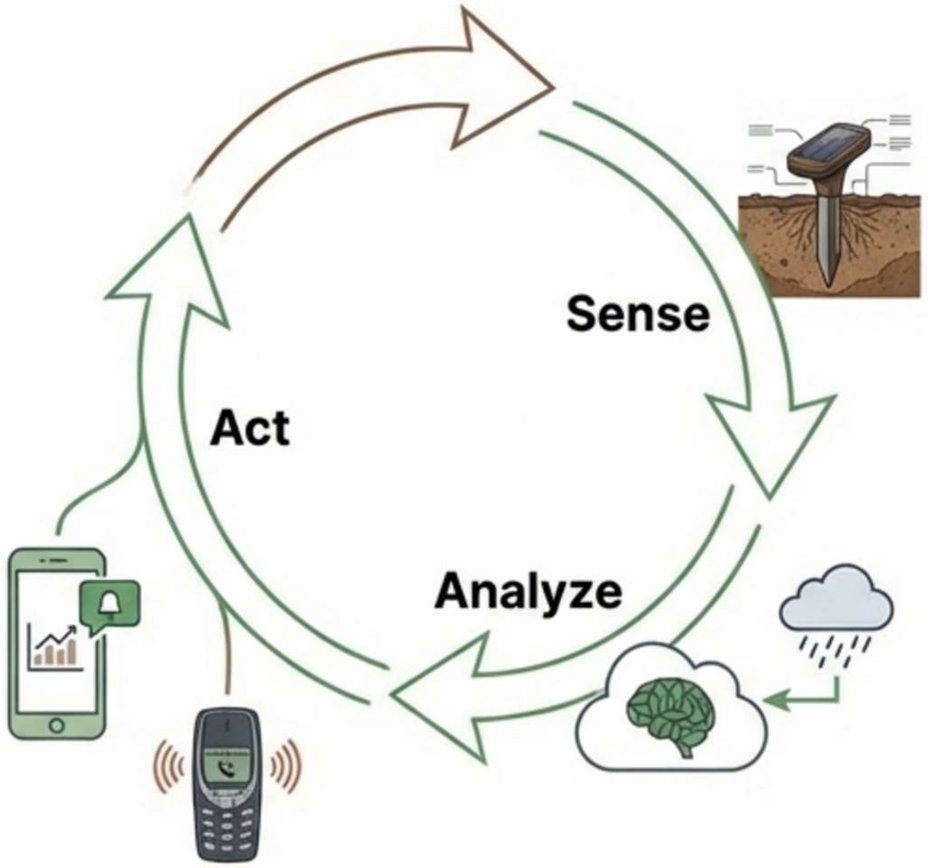
End-to-End System Architecture



The "Sense-Analyze-Act" Feedback Loop

The ‘Sense-Analyze-Act’ Loop

The system creates a closed feedback loop.
Sense: The AgroSense spike wakes from deep sleep, captures soil telemetry (moisture, temperature, humidity), and pushes it to the cloud.
Analyze: The ‘AgroMind’ AI correlates this ground truth with weather forecasts (e.g., ‘Soil is dry, but rain is predicted in 2 hours -> Suppress Alert’).
Act: The system splits the output based on the user’s tech level. Smartphone users get a push notification and visual graph; feature phone users receive an automated voice call in their local dialect.



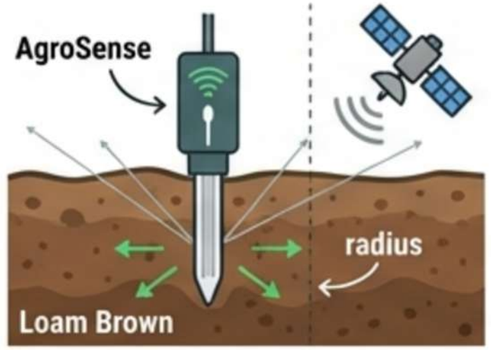
The Novelty

The ‘Pull’ Mechanism

Crucially, the system supports a ‘Pull’ feature via Kisan-Vani. Any farmer can call a toll-free number to hear a real-time status report of their field, effectively carrying their farm in their pocket.

FEATURES AND NOVELTY

Hyper-Local "Ground Truth"



Unlike competitors relying on low-resolution satellite data (20km averages), we use in-situ sensors. This captures micro-climate realities, such as specific humidity pockets that trigger fungal blight.

Neuro-Symbolic Diagnosis

We combine Computer Vision (leaf photos) with Sensor Data. A photo of a yellow leaf creates a guess; a photo *plus* soil moisture data creates a diagnosis. This eliminates false positives (e.g., distinguishing drought stress from root rot).

Frugal Engineering (< ₹800 BOM)

We achieved industrial-grade utility at a hobbyist price point. By using ESP32 deep-sleep architecture and offloading processing to the cloud, we reduced the Bill of Materials to under ₹800, compared to ₹15,000+ for commercial soil probes.

"Peace of Mind" Telemetry

The 'Kisan-Vani' feature allows for zero-touch monitoring. Farmers can leave their fields unattended, knowing they will receive a phone call if—and only if—a crisis occurs.

Ram’s Story: From worry to Action

Context: Ram spots black spots on leaves after a humid, foggy week in West Bengal.

Interaction: He opens AgroWise, speaks in Bengali describing the weather, and snaps a photo.

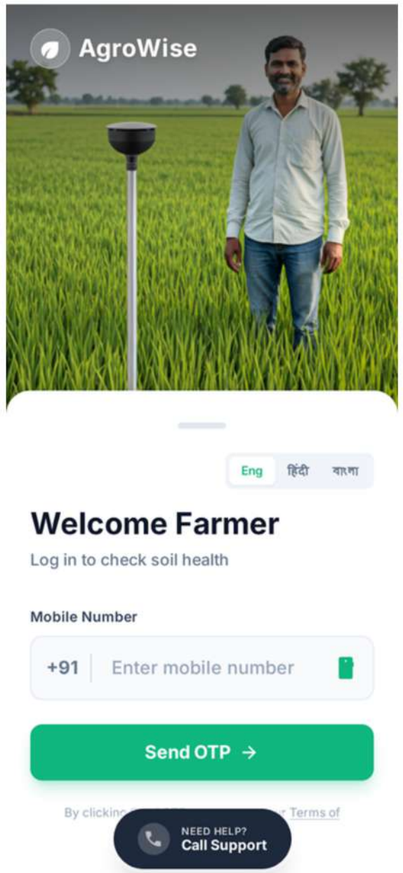
Result: The App responds immediately.

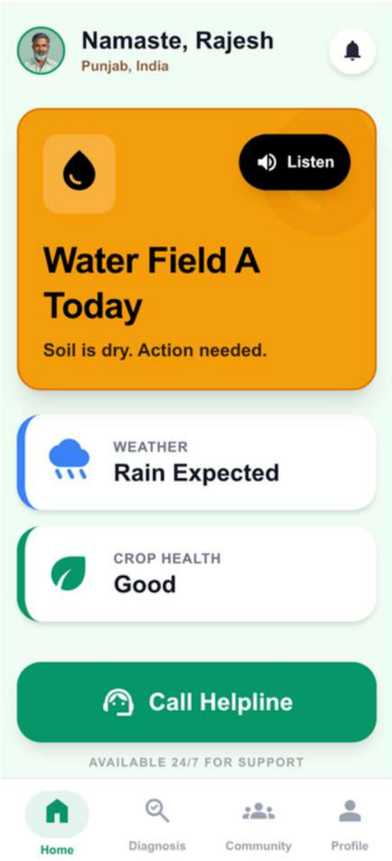
AgroWise Advice: “This is Late Blight. High humidity is spreading it. Do not water today. Spray Mancozeb immediately.""”

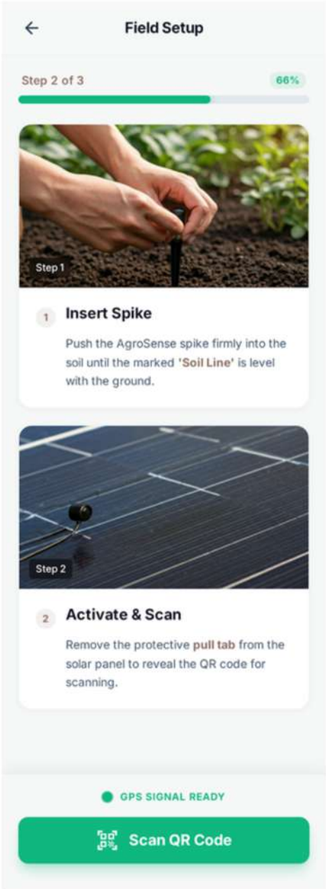
Specific diagnosis. Specific chemical recommendation. Immediate timing.

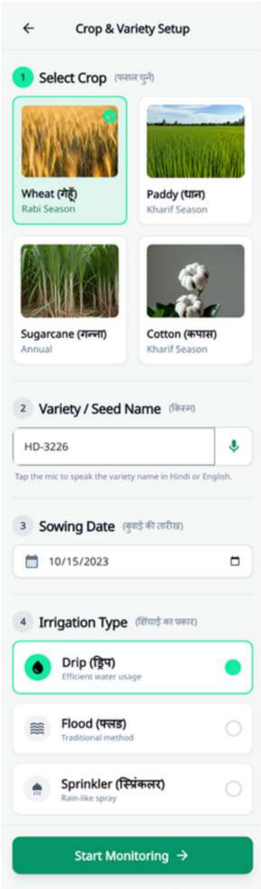
AgroWise PWA: The Farmer's Command Center

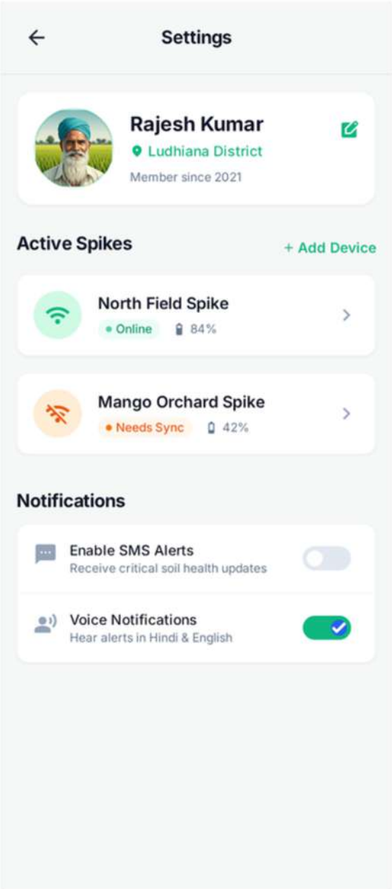
High-fidelity PWA design for seamless Device Onboarding, Real-time Monitoring, and Calibration.

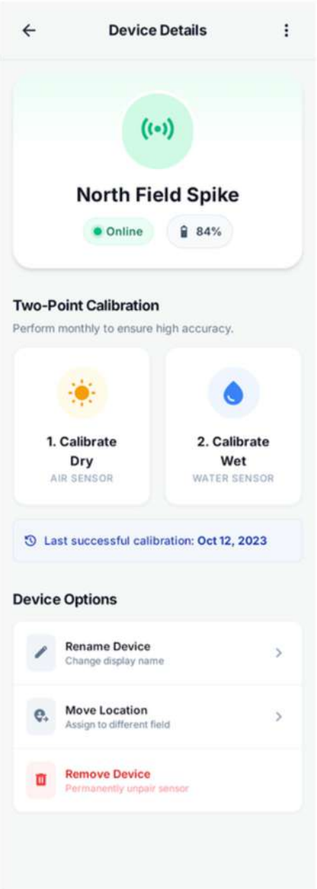












DRAWBACKS AND SHOWSTOPPERS

Risk 1: Theft & Vandalism

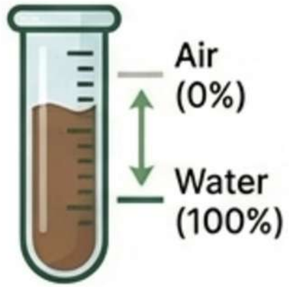
The device features a discreet 'camouflaged' design driven flush with the soil surface to minimize visibility. The ESP32 utilizes WiFi triangulation/Geofencing to trigger a 'Theft Alert' if the device moves outside the farm boundaries.

Risk 2: Connectivity Gaps

While the prototype uses GSM/WiFi, the production roadmap includes upgrading to **LoRaWAN**. This allows a single gateway to connect sensors across a 10km radius without requiring a SIM card for every device, solving the coverage issue in remote areas.

Risk 3: Soil Variability

Sensor readings vary by texture (Clay vs. Sand). We implemented a 'Two-Point Calibration' mode in the firmware, allowing farmers to capture custom 'Air' (0%) and 'Water' (100%) baselines to calibrate the device specifically for their soil type.



Team Name: NamoFans

Meet The Team

Animesh Raj

AI/ML Engineer

animeshraj958@gmail.com
+91 7004140025

Devansh Gupta

Lead Developer

devanshg308@gmail.com
+91 9927286111

Nishant Kumar

Product Manager

kumarnishant262626@gmail.com
+91 7277245100

Chirag Sharma

UI/UX Designer

chirag003214@gmail.com
+91 8302966328

THANK YOU

AgroWise & AgroSense

Democratising High-Tech Agronomy for 100% of the Farming Population.